Negar Fathi

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Profile

I am a Ph.D. candidate in Computer Science at the University of Nebraska–Lincoln. My research focuses on formal methods, program analysis, and software testing to improve software reliability and correctness. My work also explores how artificial intelligence, including large language models, can be combined with rigorous program reasoning to support software engineering.

Education

Ph.D. in Computer Science *2023–Present*
University of Nebraska–Lincoln, Lincoln, NE, USA
GPA: 3.88/4.00
Research Focus: *Applying formal methods and program analysis techniques to verify safety and liveness properties, with a particular emphasis on termination and non-termination reasoning.*
Advisor: Dr. Rahul Purandare

M.Sc. in Computer Science *2023–2026*
University of Nebraska–Lincoln, Lincoln, NE, USA
GPA: 3.94/4.00
Completed a course-based M.Sc. alongside Ph.D. studies with a focus on advanced computer science coursework.

M.Sc. in Computer Engineering (Software) *2018–2021*
Iran University of Science and Technology, Tehran, Tehran, Iran
GPA: 18.69/20.00 (*Ranked 1st*)
Thesis: *Development of a Constraint Solver to Determine the Domain for Complex Data Types*
Advisor: Dr. Saeed Parsa

B.Sc. in Computer Engineering (Software) *2013–2017*
Babol Noshirvani University of Technology, Babol, Mazandaran, Iran
GPA: 17.29/20.00 (*Ranked 1st*)
Final Project: *Study and Investigation of Routing Protocols in Wireless Sensor Networks*
Advisor: Dr. Mojtaba Mansouri

Skills

Program Verification & Analysis Tools: Roslyn, LLVM/Clang, Z3, KLEE, SeaHorn, Frama-C, DG
Software Testing & Fuzzing: AFL, NUnit, IntelliTest
Programming Languages: C/C++, C#, Python
Frontend Development: HTML, CSS, JavaScript
Backend Development: Entity Framework Core, ASP.NET Core (Web API, MVC)
Databases: Microsoft SQL Server, PostgreSQL
Systems & Development Tools: Docker, Git

Publications

- N. Fathi, H. Unno, T. Terauchi, and R. Purandare, "Sound Termination and Non-termination Analysis of C Programs with Bit-Precise Bounded Semantics and Advanced Constructs," *Proceedings of the ACM on Software Engineering*, vol. 3, no. FSE, pp. 4505–4528, Jun. 2026, doi: 10.1145/3808205.
- A. Kalaee, S. Parsa, and N. Fathi, "COSMOS: A Comprehensive Framework for Automatically Generating Domain-Oriented Test Suite," *Information and Software Technology*, vol. 154, p. 107091, Feb. 2023, doi: 10.1016/j.infsof.2022.107091.
- N. Fathi, R. Purandare, T. Terauchi, and H. Unno, "Loop-Based Slicing and Input-Driven Concretization:

An Empirical Study of Termination and Non-Termination Analysis," *arXiv preprint* arXiv:2607.08988, 2026. [Under review at *IEEE Transactions on Software Engineering*.]
- N. Fathi, "EDFNet: Early Fusion of Edge and Depth for Thin-Obstacle Segmentation in UAV Navigation," *arXiv preprint* arXiv:2604.09694, 2026. [Under review at *Third Workshop on Computer Vision for Geospatial Image Analysis, IEEE/CVF WACV*.]

Experience

Teaching Assistant *Aug. 2025–Present*
University of Nebraska–Lincoln, Lincoln, NE, USA

- Assisted in teaching undergraduate and graduate computer science courses, supported students with course material, graded assignments and exams, and held office hours.
- Courses: CSCE 322 (Programming Language Concepts); CSCE 423/823 (Design and Analysis of Algorithms); CSCE 428/828 (Automata, Computation, and Formal Languages).

Graduate Research Assistant *Aug. 2023–Dec. 2025; Jun.–Jul. 2026*
University of Nebraska–Lincoln, Lincoln, NE, USA
Advisor: Dr. Rahul Purandare

- Applied formal methods and program analysis techniques to verify safety and liveness properties, with a particular emphasis on termination and non-termination reasoning.
- Contributed to the NSF-supported project *"Collaborative Research: FMitF: Track II: Enabling Pluggable Runtime Verification Optimizations"* (NSF Award No. 2525242; PI: Dr. Rahul Purandare).

Selected Projects

- **Nexus** Developed a neuro-symbolic framework for C program termination and non-termination analysis, integrating LLVM-based loop extraction, LLM-guided witness synthesis, and SMT-based validation with iterative refinement.
- **Athena** Developed a sound framework for termination and non-termination analysis of C programs, incorporating bit-precise reasoning, bounded semantics, and support for complex language constructs.
- **FocusTNT** Developed a tool-agnostic preprocessing framework for C/C++ termination and non-termination analysis, using loop-based program slicing and input-driven concretization to simplify analysis for existing verification tools.
- **Atropos** Developed a hybrid termination and non-termination analysis framework for C/C++ programs, combining dynamic loop-entry reachability analysis with static precondition synthesis and refinement.
- **PhysIQ** Developed a hybrid rule-based and machine-learning framework for automatically detecting and classifying physical-unit inconsistencies in ROS/C++ robotic software.
- **llvm2KITTeL** Extended an LLVM-IR-to-transition-system translator with support for arrays, structures, and bit-level operations, enabling automated analysis of complex program constructs.
- **COSMOS** Developed a constraint-based test generation framework for automatically synthesizing domain-specific test inputs from program path constraints.
- **CleanCode** Developed a Roslyn-based static analysis tool for detecting violations of naming conventions, method-structure guidelines, and code-complexity rules in C# programs.
- **SoftModules** Developed a software modularization and architecture recovery tool using class-level dependency analysis to identify structural relationships and module boundaries.
- **ModelTests** Designed and tested an object-oriented C# application using interface-based design, test stubs, and NUnit-based unit testing.
- **EDFNet** Developed and evaluated an early-fusion semantic segmentation framework integrating RGB, depth, and edge information for thin-obstacle detection in UAV navigation using the DDOS dataset.
- **TaxiClusters** Developed a spatial trajectory analysis pipeline using data preprocessing and DBSCAN clustering to identify geographic patterns in large-scale taxi data.

- **WarehouseDB** Developed a warehouse management system integrating a PostgreSQL database, SQL stored procedures and functions, and a C# Windows Forms application.
- **SyncAsync** Implemented a WCF-based client-server application to compare synchronous and asynchronous communication using C# Windows Forms clients.
- **HospitalFlow** Analyzed and modeled hospital workflows to identify process inefficiencies and propose improved future-state operational processes.

Academic Service

- Ad hoc Reviewer, *Machine Intelligence Research*, 2026.

Certifications

- Oregon Programming Languages Summer School (OPLSS): Types, Logic, and Formal Methods *(2025)*; Types, Semantics, and Applications *(2024)*.
- Learning Information Technology Lab (LAITEC), Sharif University of Technology: ASP.NET MVC *(2020)*; ASP.NET *(2019)*; C# Programming *(2019)*.

Honors & Awards

- Othmer Fellowship, University of Nebraska–Lincoln, *2023–2026*
- Ranked 1st among M.Sc. students in Computer Engineering, Iran University of Science and Technology, *2021*
- Ranked 1st among B.Sc. students in Computer Engineering, Babol Noshirvani University of Technology, *2017*

Languages

Persian (Native), English (Fluent)

Last updated: October 10, 2026